Ollie Arrison

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SUMMARY

Computer graphics software engineer and 2D/3D artist focused on animation R&D, artist-facing tools, and GPU rendering. Build in C++/GLSL across Blender source development, interactive animation tools, and Unity XR.

TECHNICAL SKILLS

C++ • Python • C# • JavaScript • GLSL • PyTorch • Blender (EEVEE, Compositing, Geometry Nodes) • Vulkan • Unity • Three.js • Vue.js • OpenMP • GPU compute • real-time rendering • animation systems • Git • Linux • perf • gprof • Valgrind

EDUCATION

Carnegie Mellon University — M.S. Computer Science, expected Dec 2027; B.S. Computer Science, May 2026
Computer Graphics concentration • GPA 3.49/4.0

SELECTED PROJECTS

Non-Photorealistic Rendering and Compositing — Oct 2025–Present

C++, GLSL, Blender, EEVEE, Geometry Nodes

• Modified Blender's source and developed compositing nodes that render 3D scenes as brush strokes while preserving the lighting and colors of existing materials.
• Accelerated compositing by more than 1,000× over a CPU prototype on tested inputs with GLSL compute shaders, enabling interactive previews.
• Designed a world-space Voronoi-noise layering strategy that lets overlapping strokes form a painterly surface without full model coverage, avoiding z-fighting.

Hybrid Animation Tool — May–Aug 2023

JavaScript, Three.js, Vue.js

• Built an experimental animation tool combining hand-drawn 2D artwork with moving 3D backgrounds, exploring ways to reduce frame-by-frame drawing.
• Implemented drawing on planes, spheres, and cylinders; projected canvas artwork into 3D and created smooth camera paths from guide points.

Parallel Delaunay Triangulation — Spring 2026

C++, OpenMP, Python

• Developed geometric helpers, a Delaunay validator, and mesh-connectivity updates; helped build and refine the OpenMP parallel implementation. Replaced repeated encroachment-list cleanup with an inserted-point flag array, reducing that step from about 10% of runtime to nearly zero.
• Generated test cases and mesh visualizations for 1,000-point Gaussian-clustered, adversarially sorted, and blue-noise inputs; profiled with gprof and perf and debugged memory leaks with Valgrind. The team's parallel divide-and-conquer version achieved 5.12× speedup over its sequential baseline on 10,000 points using 32 threads on PSC's Bridges-2.

Real-Time Watercolor Renderer — Spring 2026

C++, Vulkan, GLSL, compute shaders

• Co-developed watercolor stylization for animated 3D scenes, combining a watercolor reflectance model with GPU passes for Gaussian blur, depth-aware pigment bleeding, and paper-texture compositing.
• Tuned color layering and depth-discontinuity handling to preserve silhouettes; measured frame time across filter kernel sizes.

RESEARCH EXPERIENCE

XR Researcher, Augmented Perception Lab — Carnegie Mellon University — Aug 2025–May 2026

Designed and implemented physics-based XR interfaces in Unity/C# under Professor David Lindlbauer, connecting real-time graphics engineering with human–computer interaction research.

HONORS

• Physics-Based Rendering Art Award — first place in three competitions, runner-up in one; funded trip to SIGGRAPH 2026.
• First place, Computer Graphics class showcase (2024), among approximately 70 students.
• First place, Fundamentals of Programming class final project showcase (2022), among approximately 500 students.